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MASTER'S THESIS

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**A Heuristic Approach to Routing and Spectrum Assignment
in Multi-Fiber Elastic Optical Networks: Design and
Implementation in TeraFlowSDN**

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Abstract

THESIS TITLE

MASTER'S IN COMPUTER SCIENCE AND NETWORKING

The provisioning of light-paths in flexible-grid optical networks has significantly enhanced spectrum utilization; however, the integration of topologies featuring multiple parallel links between optical devices introduces profound computational complexities in Routing and Spectrum Allocation (RSA). Specifically, when multiple transponders (muxponders) interface with the exact same degree of Reconfigurable Optical Add-Drop Multiplexer (ROADM) devices, the permutation of viable routing options increases exponentially. This geometric expansion renders standard path computation mechanisms excessively heavy and inefficient for traditional Software-Defined Networking (SDN) controllers. This routing challenge is further compounded by the stringent constraints of spectrum continuity and contiguity under the ITU-T G.694.1 standard, which dictates a 12.5 GHz minimum slot width and a high-resolution 6.25 GHz slot granularity. Because optical signals directed to the same ROADM degree cannot reuse identical frequencies without severe interference, the controller is burdened with the computationally intensive task of continuously verifying the availability of every individual 6.25 GHz slot across all potential parallel routes. To overcome these dual bottlenecks of path scalability and high-resolution spectrum mapping, this thesis introduces the `ParallelOpticalController`, a novel microservice engineered directly into the `TeraFlowSDN` ecosystem. Leveraging the `networkX` library for advanced topological graph modeling, the proposed architecture natively supports multi-directed graphs with parallel edges. This enables the highly efficient execution of Dijkstra's algorithm to compute the optimal shortest path, while concurrently discovering and suggesting exhaustive alternative path combinations to maximize network utilization. Furthermore, to resolve the heavy spectrum calculation overhead, the controller employs an innovative bitwise mapping algorithm that performs mathematical intersections of available flex-grid slots across sequential hops, drastically reducing the time required to identify contiguous spectrum blocks. Finally, these backend computational advancements are fully integrated with a dynamic graphical WebUI that visualizes the complex parallel-link topology, endpoint interfaces, and real-time spectrum allocations, thereby offering a highly scalable, end-to-end framework for next-generation optical network management.

Contents

1	Introduction	8
1.1	Overview	8
1.2	Motivation	9
1.3	Research Objectives	10
1.4	Thesis Contributions	11
1.5	Organization of the Thesis	12
2	Background and Literature Review	14
2.1	Software Defined Network	14
2.1.1	SDN Architecture and Principles	14
2.1.2	SDN in Optical Networks	15
2.2	TeraflowSDN	15
2.2.1	Architecture and Ecosystem	15
2.2.2	Key services	16
2.3	Optical Transport Networks	17
2.3.1	WDM technology	17
2.3.2	ITU standards	17
2.3.3	Network elements (devices)	18
2.3.4	Data models (configuration)	19
2.4	Routing and Spectrum Allocation	20
2.4.1	Spectrum continuity and contiguity constraints	20
2.4.2	Existing RSA algorithms	21
2.5	Problems with Parallel Links in RSA	22
2.5.1	Problem statement	22
2.6	Current approaches and limitation	23
2.7	Related work	24
3	Problem Definition and Proposed Algorithm	25
3.1	Formal Problem Statement	25
3.2	Graph Modelling with Parallel Links (topologies)	26
3.3	Proposed RSA Algorithm	27
3.3.1	Bandwidth & Slot Computation	28
3.3.2	Path Discovery	28
3.3.3	Bitmap-Based Spectrum Allocation	29
3.3.4	Slot Acquisition & Assignment	30
3.4	Complexity Analysis	31

4	System Design and Implementation	33
4.1	Overall System Architecture	33
4.2	Modifications to TeraFlowSDN	33
4.2.1	Device Service Enhancements (OCDriver, OpenROADMDriver)	33
4.2.2	Context Service & Database Schema Extensions	34
4.3	RSA Computation Module (RSATools, ITUStandards)	34
4.4	Parallel Optical Controller Microservice	35
4.4.1	Service Architecture & API Design	35
4.4.2	Graph Construction with NetworkX	35
4.5	Integration with Existing TFS Services	35
4.6	Topology & Link Management (Parallel Link Support)	35
4.7	WebUI Enhancements (Spectrum Visualization)	36
5	Testbed Setup and Deployment	37
5.1	Physical/Virtual Infrastructure	37
5.2	Emulated Devices (Transponders, ROADMS)	38
5.3	Network Topology Configuration (tfs descriptors)	38
5.4	Deployment Procedure (TeraFlowSDN & ParallelOpticalController) .	39
6	Results and Evaluation	40
6.1	Evaluation Methodology & Test Scenarios	40
6.2	Functional Validation	40
6.3	Computational Performance	41
6.4	Spectrum Utilization Efficiency	41
6.5	Throughput Analysis	41
6.6	Comparison with Baseline/Existing Approaches	42
6.7	Statistical Analysis	42
6.8	Discussion of Results	42
7	Conclusion and Future Work	43
7.1	Summary of Contributions	43
7.2	Limitations	44
7.3	Future Work Directions	44

List of Figures

1.1	Number of internet users over the time	8
1.2	An example of use of flexible the grid	9
2.1	TeraflowSDN Architecture	16
2.2	DWDM Flex-Grid Overview	17
2.3	An example optical network	19
3.1	RSA operation using ParallelOpticalController in TeraFlowSDN . . .	27

List of Tables

3.1 Efficiency evaluation of ParallelOpticalController 31

1 Introduction

1.1 Overview

The unprecedented surge in global data traffic, driven largely by the widespread deployment of 5G cellular networks and Fiber-to-the-Home (FTTH) access architectures, has fundamentally reshaped the telecommunications landscape. To satisfy this monumental and continuous demand without incurring the prohibitive costs of laying new physical fiber, network operators are compelled to maximize the transmission capacity of existing optical resources. This is primarily achieved by aggregating multiple communication channels into a single optical fiber using Wavelength Division Multiplexing (WDM) and, more specifically, Dense Wavelength Division Multiplexing (DWDM), which increases fiber efficiency by transmitting independent signals at different wavelengths. However, as individual fiber capacity nears its theoretical

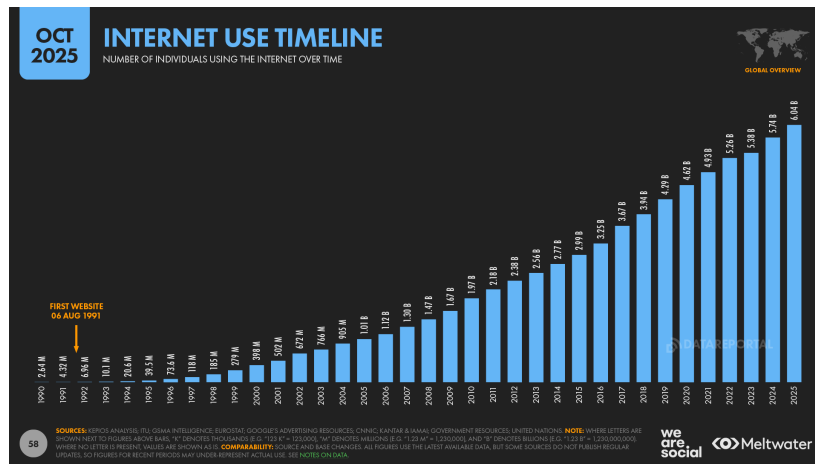


Figure 1.1: Number of internet users over the time

Shannon limits, spectral multiplexing alone is often insufficient to satisfy massive localized traffic requirements. Consequently, network operators increasingly deploy "multifiber" or parallel-link configurations—utilizing multiple physical fiber pairs between adjacent nodes—to achieve a multiplicative increase in total throughput. While protocols like Link Aggregation (LAG) are traditionally utilized at higher network layers to duplex or group these connections for redundancy and increased bandwidth, managing these resources at the optical layer introduces significant complexity.

To handle the management burden of these high-capacity, multifiber infrastructures, the optical control plane has transitioned from static, manual configurations to

highly automated, centralized management systems empowered by Software-Defined Networking (SDN). As DWDM technologies have matured, the traditional fixed-grid spectrum allocation—which assigns rigid, uniform channel spacing—has become a structural bottleneck. To overcome this, flexible-grid (flex-grid) architectures have emerged as the industry standard. By adhering to ITU-T G.694.1 standards, which utilize a precise 6.25 GHz slot granularity with 12.5 GHz slot width, flex-grid networks allow for elastic, demand-driven bandwidth allocation that vastly improves overall spectral efficiency [1]. The deployment of flex-grid systems in long-haul backbone

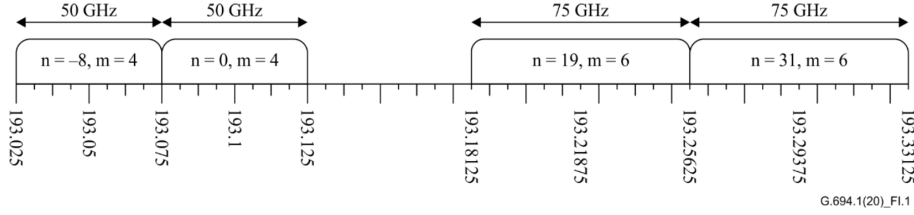


Figure 1.2: An example of use of flexible the grid

communications introduces substantial algorithmic complexity. In multi-hop networks, an optical signal must maintain both spectrum continuity—utilizing the same frequency across all links—and spectrum contiguity—utilizing adjacent frequency slots—from source to destination. This Routing and Spectrum Allocation (RSA) challenge is critically exacerbated in topologies featuring multiple parallel optical links between network elements, such as adjacent Reconfigurable Optical Add-Drop Multiplexers (ROADMs). When parallel links are introduced to a multi-hop architecture, the number of viable routing permutations increases exponentially. Consequently, traditional SDN controllers struggle to efficiently evaluate the availability of every 6.25 GHz slot across all possible path combinations, leading to severe computational bottlenecks and delayed provisioning times.

To resolve this critical scalability issue, this thesis proposes a novel approach by introducing the Parallel Optical Controller, a stateful microservice integrated directly into the TeraFlowSDN ecosystem. By leveraging advanced graph modeling techniques and a highly optimized bitwise spectrum intersection algorithm, this proposed method efficiently isolates available spectrum blocks across complex parallel routes. Ultimately, this solution effectively neutralizes the parallel link bottleneck, ensuring rapid, reliable, and highly scalable lightpath provisioning for next-generation optical networks.

1.2 Motivation

While extensive research has been conducted on Routing and Spectrum Allocation (RSA) in Elastic Optical Networks (EONs)[2][3], particularly concerning complex multi-hop lightpath provisioning[4]—there remains a notable gap in the literature regarding the efficient management of multiple parallel optical links between adjacent

devices. In traditional centralized control planes, such as those managed by other SDNs, parallel links are typically treated as isolated, individual entities rather than aggregated resources[5]. Consequently, when a lightpath request is processed, the RSA algorithm is forced to evaluate the spectrum continuity of each parallel link independently[3]. This legacy approach is highly inefficient, as it triggers redundant, computationally heavy, back-and-forth calculations that severely degrade the controller’s performance and scalability in dense optical topologies. The primary motivation of this thesis is to fundamentally resolve this parallel link computational bottleneck within the emerging TeraFlowSDN ecosystem [6]. By addressing relative algorithmic inefficiency, the research aims to design a system capable of evaluating multiple parallel paths concurrently, significantly reducing the computational overhead required to find available flex-grid spectrum.

Furthermore, theoretical RSA optimization cannot be solved just as a math problem or a simulation using static data; its efficacy is strictly dependent on the acquisition of precise, real-time telemetry from the underlying physical layer. While modern optical networks are rapidly transitioning toward vendor-agnostic architectures unified by standardized data models such as OpenConfig[7] and OpenROADM[8], a significant barrier remains in the practical deployment of centralized algorithms: hardware heterogeneity. Although these industry standards define common data models using YANG, individual equipment vendors frequently implement them with subtle variations in configuration hierarchies and telemetry reporting. Consequently, extracting uniform network state information—such as operational frequency ranges and transport capabilities—via a singular filtering procedure is often unfeasible in multi-vendor environments. This research addresses this disparity by bridging the gap between abstract path computation and physical hardware limitations through the extensive enhancement of device drivers within the TeraFlowSDN ecosystem. By refining the default driver logic and implementing a tiered approach that utilizes both standardized procedures and specialized, device-based handlers, this thesis establishes a more robust framework for multi-vendor support. These enhancements ensure that the proposed RSA algorithm is fed by accurate, model-driven telemetry data, effectively reconciling high-level computational logic with the practical complexities of diverse optical hardware management.

1.3 Research Objectives

To systematically address the challenges of Routing and Spectrum Allocation (RSA) over parallel optical links, this thesis is guided by the following core research objectives:

- **Analysis of Flexible-Grid Standards:** To investigate the operational principles of flexible-grid optical networks, specifically adhering to the ITU-T G.694.1 standards regarding spectrum slot width and granularity.

- **Evaluation of SDN Optical Intent Provisioning:** To study how optical intents and lightpath requests are currently handled by popular, industry-standard Software-Defined Networking (SDN) controllers, identifying their limitations in dense, multi-link topologies.
- **Examination of TeraFlowSDN Device Drivers:** To conduct a comprehensive study of the telemetry and configuration mechanisms within TeraFlowSDN, with a specific focus on analyzing the OpenConfig and OpenROADM device drivers.
- **Assessment of the Existing Optical Controller:** To critically evaluate the official OpticalController microservice currently operating within the TeraFlowSDN ecosystem, identifying its architectural constraints when processing complex optical intents.
- **Review of RSA Algorithmic Approaches:** To explore and analyze contemporary algorithmic strategies used to solve the RSA problem in multi-hop, multi-fiber optical networks, evaluating their computational efficiency and spectrum assignment mechanisms.
- **Formulation of an Optimal Solution:** To synthesize the findings from the aforementioned studies and propose an optimal, highly scalable RSA framework that efficiently resolves the computational bottlenecks associated with parallel optical links.

1.4 Thesis Contributions

Building upon the established research objectives, this thesis makes several tangible contributions to the field of optical Software-Defined Networking (SDN), specifically addressing the computational bottlenecks associated with routing and spectrum assignment over parallel links. The core contributions of this work are summarized as follows:

- **Enhancement of Optical Device Drivers:** A significant extension of the existing TeraFlowSDN device drivers, specifically the OCDriver (for OpenConfig devices) and the OpenROADM driver. This contribution includes the implementation of custom filters to extract vendor-agnostic telemetry—such as minimum and maximum operating frequencies, transport types (e.g., OCH, OMS), and DWDM parameters—directly from XML configurations, seamlessly integrating this data into an extended Context database schema.
- **Design of an Optimized RSA Algorithm:** The formulation and implementation of a highly scalable Routing and Spectrum Allocation (RSA) algorithm tailored for topologies with multiple parallel links. This includes utilizing the

`networkX` library to construct multi-directed graphs that accurately represent parallel edges, alongside the introduction of a novel bitmap-based bitwise intersection method designed to rapidly compute the availability of contiguous spectrum slots across multiple hops respecting the parallel links.

- **Development of the ParallelOpticalController Microservice:** The primary contribution of this thesis is the design, implementation, and seamless integration of a novel, stateful microservice—the ParallelOpticalController—into the TeraFlowSDN ecosystem, which serves as the practical execution engine for the proposed Routing and Spectrum Allocation (RSA) algorithm. This implementation is explicitly engineered to navigate multi-hop topologies featuring parallel optical links between nodes at each hop, whereas traditional controllers often struggle with the exponential increase in viable routing permutations. The controller autonomously calculates paths that strictly adhere to both spectrum continuity and contiguity constraints, ensuring valid end-to-end lightpath establishment across disjoint physical resources while utilizing optimized logic to mitigate the computational bottlenecks associated with solving NP-hard routing problems in large-scale networks. By facilitating the high-speed discovery of primary shortest paths and evaluating alternative route combinations in real-time, the architecture provides robust support for dynamic traffic demands and enhances overall network resilience.
- **Advanced Network Visualization and WebUI Integration:** To comprehensively upgrade the TeraFlowSDN Web User Interface (WebUI) to reflect the complex physical and logical states of the optical network. This objective focuses on developing visual representations of device endpoint capabilities, real-time spectrum slot availability (based on ITU-T 6.25 GHz granularity), multi-hop topological links, and the dynamic output of the RSA computation.

By fulfilling these objectives, this research aims to deliver an end-to-end solution that spans from low-level physical device configuration up to high-level algorithmic path computation and graphical user interaction.

1.5 Organization of the Thesis

The remainder of this thesis is logically structured to guide the reader through the theoretical background, the proposed algorithmic solutions, the system implementation, and the final evaluation. The document is organized as follows:

- **Chapter 2:** Background and Literature Review provides the foundational concepts necessary to understand the research. It discusses the principles of Software-Defined Networking (SDN) in optical networks, introduces the TeraFlowSDN architecture, and details Optical Transport Networks (OTN)

including WDM technology and ITU standards. The chapter concludes with a review of existing Routing and Spectrum Allocation (RSA) algorithms, highlighting the specific problems and limitations associated with parallel links.

- **Chapter 3:** Problem Definition and Proposed Algorithm formally establishes the research problem and introduces the mathematical formulation of the challenge. It details the graph modeling techniques used to represent topologies with parallel links and breaks down the proposed RSA algorithm, covering bandwidth computation, path discovery, bitmap-based spectrum allocation, and slot acquisition, followed by a thorough complexity analysis.
- **Chapter 4:** System Design and Implementation delves into the practical engineering and software architecture of the proposed solution. It outlines the modifications made to TeraFlowSDN—including device driver enhancements for OpenConfig and OpenROADM YANG models, database schema extensions, and the RSA Computation Module. Furthermore, it details the development of the new Parallel Optical Controller microservice and the corresponding WebUI enhancements for spectrum visualization.
- **Chapter 5:** Testbed Setup and Deployment outlines the experimental environment used to validate the proposed system. It describes the physical and virtual infrastructure, the configuration of emulated transponders and ROADM devices, the setup of network topologies, and the deployment procedure utilizing MicroK8s.
- **Chapter 6:** Results and Evaluation presents a comprehensive analysis of the system's performance. It covers functional validation, computational performance, and spectrum utilization efficiency. The chapter also provides a throughput analysis, a comparative study against baseline approaches, and a statistical discussion of the results.
- **Chapter 7:** Conclusion and Future Work summarizes the primary contributions of the thesis, discusses the current limitations of the implemented system, and suggests potential directions for future research and development.

2 Background and Literature Review

This chapter establishes the theoretical foundation and contextual background for the research presented in this thesis, tracing the technological progression from foundational Optical Transport Networks (OTN) to modern automated control architectures. The review begins by examining the evolution of Wavelength Division Multiplexing (WDM), detailing the industry-wide transition from rigid, fixed-grid spectrum allocation to the highly efficient, elastic paradigms of flexible-grid (flex-grid) EONs. It contrasts traditional, decentralized optical control planes with the emergence of Software-Defined Networking (SDN), which provides the centralized intelligence required for dynamic resource management. Within this context, the chapter details the architectural components of contemporary SDN controllers, specifically focusing on the TeraFlowSDN ecosystem. A critical analysis of current Routing and Spectrum Allocation (RSA) methodologies follows, identifying a persistent research gap: the inability of traditional path computation and spectrum assignment logic to scale effectively when encountering multi-hop topologies with parallel optical links. Finally, the chapter introduces the `ParallelOpticalController` as a transformative solution within TeraFlowSDN, designed to resolve these computational bottlenecks and provide high-speed, constraint-aware provisioning for complex multifiber infrastructures.

2.1 Software Defined Network

2.1.1 SDN Architecture and Principles

Software-Defined Networking (SDN) represents a fundamental paradigm shift in telecommunications, decoupling the network’s control logic—the control plane—from the underlying packet or optical forwarding hardware, known as the data plane. This structural separation centralizes network intelligence into a software-based controller that maintains a global, real-time view of the network topology and its available resources. By abstracting the physical infrastructure through open, standardized Southbound Interfaces (SBIs) such as NETCONF or gNMI, SDN enables deep network programmability and dynamic orchestration of forwarding behaviors. This centralized management framework allows high-level applications to request network resources through Northbound Interfaces (NBIs) without requiring granular knowledge of the physical layer’s complexity. Consequently, SDN provides the necessary architecture to achieve multi-vendor interoperability and agile service

provisioning, transforming rigid physical infrastructure into flexible, automated, and software-governed assets.

2.1.2 SDN in Optical Networks

While initially designed for packet-switched networks, SDN principles have been aggressively adapted for the optical domain to create Software-Defined Optical Networks (SDON). Unlike packet networks, optical networks must manage analog physical layer impairments, wavelength continuity, and complex spectrum allocation constraints. SDON abstracts these physical layer complexities by decoupling the intelligent control logic—termed Optics Defining Software (ODS)—from the programmable physical hardware, known as Software Defined Optics (SDO) [9]. This abstraction allows the central controller to dynamically provision lightpaths, tune variable transponders for specific modulation formats, and automatically configure advanced Reconfigurable Optical Add-Drop Multiplexers (ROADMs), successfully moving away from historically static, manually operated optical grids. To manage this complexity at scale, modern SDON deployments frequently adopt a Transport-SDN (T-SDN) architecture, this is often achieved through a hierarchical control structure [10]. In this model, lower-level controllers manage specific optical nodes, while a higher-level controller (such as ONOS) facilitates multi-vendor integration using standardized interfaces like the Transport API (T-API) over NETCONF and RESTCONF protocols. Furthermore, breaking vendor lock-in is a critical focus of modern optical control planes and initiatives like the Open Disaggregated Transport Network (ODTN) utilize standardized YANG data models (such as OpenConfig and OpenROADM) to abstract heterogeneous hardware [11]. This standardization enables the seamless, end-to-end integration of packet and optical layers across disaggregated white-box devices. Finally, as emphasized in "Network Slicing in SDN Networks," this centralized, programmable control is what ultimately enables advanced network virtualization. By leveraging SDON capabilities, operators can deploy multiple isolated logical slices over a single shared physical infrastructure, dynamically shaping traffic to guarantee specific Quality of Service (QoS) and performance isolation for varying applications.

2.2 TeraflowSDN

2.2.1 Architecture and Ecosystem

TeraFlowSDN (TFS) is a state-of-the-art, open-source SDN controller developed under the European Telecommunications Standards Institute (ETSI) to radically advance the orchestration of beyond 5G and 6G networks. Designed as a secure, cloud-native platform, TFS employs a microservices-based architecture orchestrated within Kubernetes environments, enabling individual control components to oper-

ate autonomously and scale independently. This inherent modularity provides an ideal ecosystem for integrating custom logic, such as vendor-agnostic device drivers and novel path computation engines. Furthermore, the architecture is explicitly engineered to seamlessly integrate with modern Network Function Virtualization (NFV) and Multi-access Edge Computing (MEC) frameworks, including ETSI Open-SourceMANO. By unifying network and cloud resource management, TeraFlowSDN equips telecom operators and hyperscale cloud providers with zero-touch automation, high business agility, and dynamic service provisioning across complex, multi-tenant optical and packet transport infrastructures.

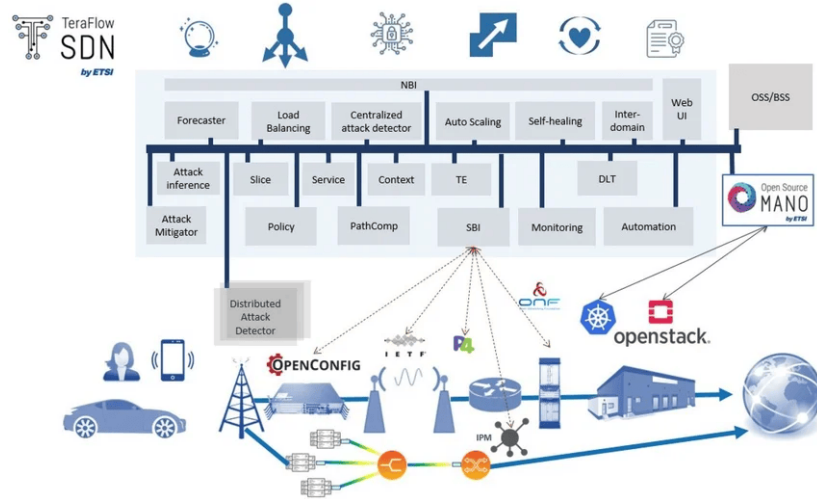


Figure 2.1: TeraflowSDN Architecture

2.2.2 Key services

The TeraFlowSDN ecosystem relies on a highly coordinated suite of specialized microservices, communicating via Remote Procedure Calls (gRPC), to maintain network state and execute dynamic configurations. At the core of this architecture is the Context Service, which acts as the highly available, stateful storage mechanism maintaining the definitive global view of multi-domain topologies, devices, endpoints, and active network slices. Interacting closely with this centralized database is the Device Service, which manages all southbound interfaces by utilizing a modular framework of vendor-agnostic drivers—often leveraging NETCONF and OpenConfig models—to dynamically onboard and configure physical or emulated hardware. Furthermore, the Service Component oversees the end-to-end lifecycle of network connections by interpreting incoming connectivity requests and delegating cross-layer configurations to appropriate handlers. For optical transport infrastructures specifically, the OpticalController operates as a dedicated service to process complex optical intents, executing resource allocation algorithms such as Routing and Spectrum Assignment (RSA) to establish continuous and contiguous lightpaths across multi-hop networks. These fundamental operations are visually managed through a graphical WebUI that provides topology visualization and telemetry display, while

the broader ecosystem is fortified by complementary microservices dedicated to machine learning-based security, distributed ledger technologies for multi-tenancy, and policy-driven automation to ensure robust, zero-touch network orchestration.

2.3 Optical Transport Networks

2.3.1 WDM technology

Wavelength Division Multiplexing (WDM), specifically Dense WDM (DWDM), is the foundational technology for modern high-capacity optical networks. DWDM enables the transmission of multiple data streams over a single optical fiber by modulating each stream onto a distinct wavelength (or color) of laser light. Historically, DWDM utilized a fixed-grid spacing, allocating rigid channel intervals of 50 GHz, 100 GHz, or even wider. However, as data rates increased and traffic patterns became more unpredictable, this rigid spacing resulted in inefficient spectrum utilization. To address this bottleneck, modern elastic optical networks (EONs) utilize flexible-grid (flex-grid) technologies[12]. As defined in the ITU-T G.694.1 standard, the flex-grid allows a mixed bit rate or mixed modulation format transmission system to allocate frequency slots with different widths, optimizing the spectrum for the specific bandwidth requirements of individual channels[1]. This transition from fixed to flexible allocation is what necessitates advanced Routing and Spectrum Allocation (RSA) algorithms, such as the one proposed in this thesis, to dynamically manage the resulting variable-width spectrum blocks across complex, multi-fiber topologies.

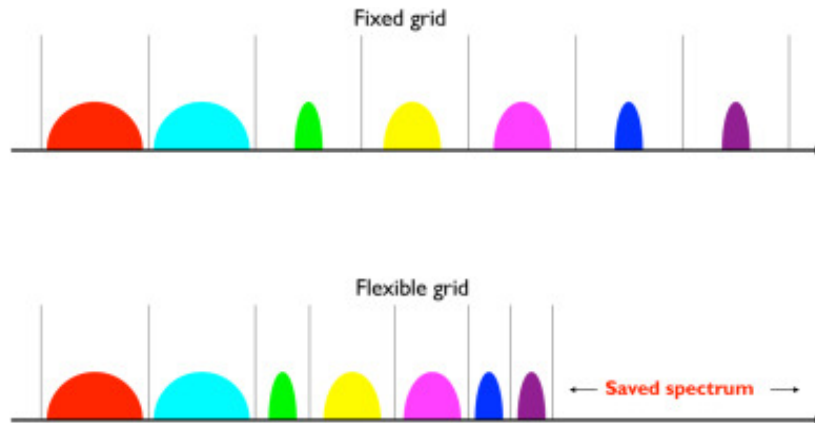


Figure 2.2: DWDM Flex-Grid Overview

2.3.2 ITU standards

The implementation of these flex-grid networks is strictly governed by International Telecommunication Union (ITU) standards to ensure interoperability and consistent spectrum management. Specifically, the ITU-T G.694.1 Recommendation provides the definitive frequency grid for DWDM applications. Under this standard, the

entire DWDM spectrum is anchored to a reference central frequency of 193.1 THz. To enable elastic allocation, the standard defines a "frequency slot" as a contiguous frequency range characterized by its nominal central frequency and its total slot width. The architecture is built upon two critical granularities:

- **Nominal Central Frequency Granularity:** The allowed central frequencies are defined by the formula $193.1 + n \times 0.00625$ THz, establishing a fine-grained granularity of 6.25 GHz to allow precise positioning of channels.
- **Slot Width Granularity:** The total width of any assigned frequency slot is defined by $12.5 \times m$ GHz, establishing 12.5 GHz as the minimum, indivisible building block for bandwidth allocation.

2.3.3 Network elements (devices)

The physical topology comprises two primary categories of optical nodes: terminal multiplexers (transponders and muxponders) and Reconfigurable Optical Add-Drop Multiplexers (ROADMs). In modern high-capacity optical networks, network end-points are more precisely characterized as muxponders, which aggregate multiple lower-rate client-layer signals into high-speed optical channels suitable for DWDM transmission. These devices utilize bidirectional ports equipped with advanced pluggable transceivers that handle complex modulation and Forward Error Correction (FEC). While the telecommunications industry employs a diverse array of transceiver form factors and optical modules—such as QSFP, QSFP28, OpenZR+, and 400ZR—each offering varying frequency band support, capacity, and error correction capabilities, this thesis adopts a specific, standardized hardware profile for its emulation environment. Specifically, the emulated muxponder ports are modeled using the QSFP56-DD form factor equipped with Digital Coherent Optics, utilizing a 400GBASE-ZR Ethernet Physical Medium Dependent (PMD) sublayer and automated FEC. Although real-world deployments may feature heterogeneous transceiver capabilities, a foundational assumption of this research is that all muxponder ports are fully bidirectional and provide continuous, tunable frequency support across the entire C-band. This baseline ensures a uniform spectrum allocation space for evaluating the proposed RSA algorithms. Conversely, ROADMs serve as the optical switching backbone, enabling individual wavelengths to be dynamically added, dropped, or passed through a node entirely in the optical domain, thereby bypassing costly optical-to-electrical-to-optical (O-E-O) conversions. To accurately evaluate the Routing and Spectrum Allocation (RSA) algorithm on parallel links, it is necessary to establish strict assumptions regarding the internal architecture of these ROADMs. A typical ROADM architecture consists of Shared Risk Groups (SRGs) for local add/drop functionality and Direction Degrees (DEGs) equipped with Wavelength Selective Switches (WSS) for line-side routing. In the context of this thesis, physical fiber connections are only classified as strict "parallel links" if the ports originating from the same muxponder terminate at the same SRG and are routed outward

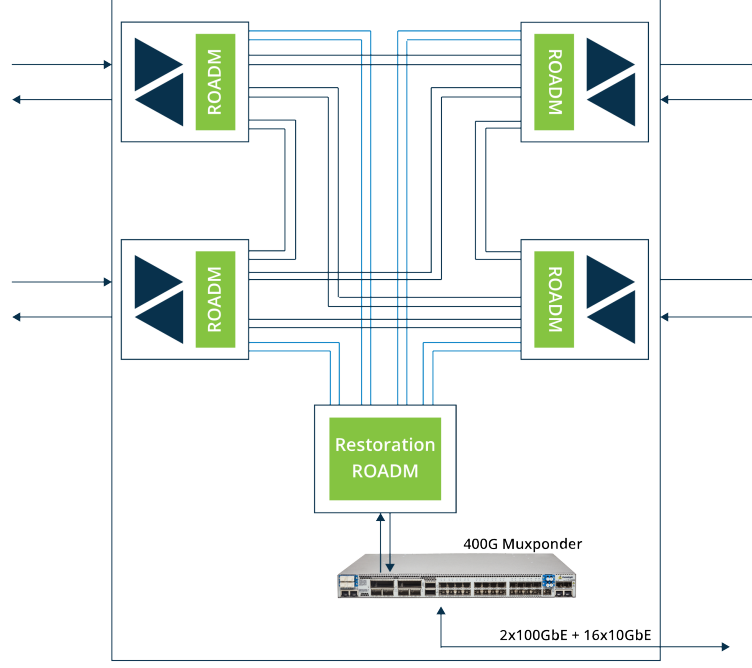


Figure 2.3: An example optical network

through the exact same DEG of the ROADM. Under this specific topological constraint, the parallel fibers share an identical physical routing vector, making spectrum contiguity and continuity algorithms highly complex, as the spatial routing does not inherently resolve spectrum overlap. If the ports were to route through different DEGs, the spatial separation provided by distinct WSS hardware would allow for simple frequency re-use, thereby negating the specific parallel-link RSA constraints addressed in this study. Furthermore, to accommodate high-capacity throughput and scalability, the ROADM devices in this simulated environment are assumed to be wideband-capable, operating seamlessly across the entire S, C, and L (SCL) optical bands.

2.3.4 Data models (configuration)

To achieve true interoperability and break the traditional hardware lock-in imposed by proprietary management interfaces, modern optical equipment relies on standardized YANG data models. OpenConfig[7] and OpenROADM[8] have emerged as the two prominent industry standards used to formalize the representation of optical device configurations and operational states. By structuring parameters in universally understood XML or JSON formats, these models hide the underlying hardware idiosyncrasies from the control plane, allowing an SDN controller to parse telemetry and push configurations uniformly regardless of the equipment manufacturer. To manage the complexity of optical networking, these data models utilize strictly defined XML namespaces to isolate specific operational domains. Within the emulation environment of this thesis, the terminal devices (muxponders) primarily utilize the

OpenConfig standard. For instance, the physical hardware hierarchy—such as form factors and transceivers—is defined under the [plaform-namespace](#). Concurrently, the operational optics are managed via the [terminal-device namespace](#), which dictates precise parameters such as operational frequencies (e.g., 191.6 THz), target output power, and the logical channel assignments mapping Ethernet client traffic to Optical Transport Network (OTN) line ports.

Conversely, the emulated optical switches utilize the OpenROADM data model, which is specifically tailored to describe complex photonic switching architectures. Using namespaces such as [device namespace](#) and [interface namespace](#), the model accurately maps the internal ROADM topology. It defines critical switching components including circuit packs (such as twin Wavelength Selective Switches), Direction Degrees (DEGs), Shared Risk Groups (SRGs), and specific optical interfaces like Optical Transport Section (OTS) and Optical Multiplex Section (OMS). By adhering to these standardized schemas, the SDN controller is decoupled from vendor-specific command-line interfaces. Consequently, network operators can seamlessly deploy heterogeneous, multi-vendor optical topologies. A single controller can dynamically orchestrate an end-to-end lightpath by pushing standard OpenConfig XML payloads to tune a Cisco transceiver, while simultaneously pushing OpenROADM payloads to configure cross-connects on an Ericsson ROADM, effectively unifying the entire physical layer.

2.4 Routing and Spectrum Allocation

2.4.1 Spectrum continuity and contiguity constraints

Establishing a lightpath in an Elastic Optical Network (EON) requires solving the complex Routing and Spectrum Allocation (RSA) problem. In traditional fixed-grid DWDM networks, resource allocation—known as Routing and Wavelength Assignment (RWA)—was primarily concerned with satisfying a simple wavelength continuity requirement [13]. However, the evolution to flexible-grid EONs introduces highly restrictive dual constraints that transform resource allocation into an NP-hard computational challenge [13]. Regardless of the specific allocation policy utilized, every valid assignment in an EON must fundamentally satisfy three primary physical constraints [2]:

- **Spectrum Continuity:** This constraint dictates that the exact same frequency spectrum (the specific index of the allocated spectrum slots) must be utilized and remain identical across every single link along the end-to-end routing path [14]. However, in multifiber infrastructures featuring parallel links, this constraint presents a crucial operational nuance [3]. While the frequency index must remain identical end-to-end, the optical signal is permitted to shift spatially between parallel physical fibers at each hop, provided there are no internal switching restrictions at the optical cross-connects [3].

- **Spectrum Contiguity:** The frequency slots assigned to a single lightpath must be strictly adjacent to one another within the frequency domain to form a contiguous "Slot Block". In modern implementations, algorithms treat individual frequency slots as fixed at a minimum width (typically 12.5 GHz) and group them contiguously to precisely match the variable bandwidth demands of the traffic [2].
- **Spectrum Non-Overlapping:** To prevent signal interference and data corruption, this constraint ensures that any specific frequency slot on a given physical fiber is allocated to a maximum of one active lightpath at a time [3].

The tight coupling of the continuity and contiguity constraints often leads to severe spectrum fragmentation. Because of the strict contiguity requirement, having enough total available spectrum on a path is insufficient if those slots are scattered and fragmented. To enforce these rules without blocking new requests, advanced RSA algorithms manage both constraints simultaneously by treating the network state as a unified "allocation matrix". This allows the controller to search for the first available block of contiguous slots that spans uniformly across all physical edges along the chosen multi-hop route [15].

2.4.2 Existing RSA algorithms

As established, the strict enforcement of spectrum continuity and contiguity transforms network resource allocation into a highly restrictive, NP-hard computational challenge[13]. To prevent severe spectrum fragmentation and the unfair blocking of high-capacity requests, intense research has been dedicated to developing sophisticated Routing and Spectrum Allocation (RSA) algorithms[16]. Over time, these algorithms have evolved from basic wavelength assignment models into complex, multi-metric optimization strategies designed to maximize network throughput and accommodate diverse, dynamic traffic demands[2]. Early approaches sought to decouple the routing and assignment phases to mitigate computational bottlenecks. Foundational algorithms employed greedy heuristics and K-shortest paths to maximize single-hop traffic while strictly enforcing standard wavelength continuity[4]. As networks transitioned to flex-grid EONs, researchers introduced offline traffic sorting heuristics to groom traffic more efficiently. For instance, the Minimum Hops with Least Slots (MHLS) algorithm prioritizes connection requests by calculating a sorting parameter[13]:

$$NP = Hops \times Slots$$

By serving "smaller" requests first, the algorithm minimizes overall blocked traffic[13]. More dynamic approaches have integrated custom cost functions to balance multiple network metrics simultaneously. For example, the FRSO-RSA algorithm evaluates candidate alternative routes by actively steering traffic away from congested or

unreliable paths using the following cost function[17]:

$$cost_c = \alpha \times PF_c + (1 - \alpha)OU_c$$

where PF_c represents the physical failure rate of the path, OU_c denotes the spectrum occupancy rate, and α acts as an adjustable weight parameter[17]. To combat the specific issue of contiguity-induced fragmentation, recent literature has shifted toward "next-state-aware" algorithms. Metrics such as Maximum Spectrum Completeness (MSC) are used to evaluate how a current assignment will impact future capacity, actively selecting routes that preserve large, unbroken blocks of spectrum [18]. Other strategies introduce highly dynamic link-weighting mechanisms that actively draw traffic toward links harboring the largest unfragmented spectrum slices rather than just looking at total available capacity [19]. When fragmentation inevitably occurs during the network's lifecycle, complex defragmentation algorithms utilizing Integer Linear Programming (ILP) and Resource Dependency Digraphs (RDD) are deployed to mathematically recalculate and shift existing lightpaths without altering their physical routes[20]. Furthermore, advanced meta-heuristics like Genetic Algorithms and probabilistic models like Markov Decision Processes (MDP) have been explored to navigate massive solution spaces and predict network states[21], [22].

2.5 Problems with Parallel Links in RSA

2.5.1 Problem statement

While the Routing and Spectrum Allocation (RSA) problem in optical networks has been extensively studied for years, the comprehensive integration of multi-fiber or parallel links between adjacent nodes remains an underexplored area. Technologies such as link aggregation and bandwidth duplexing have become increasingly common, introducing multiple physical connections between devices to meet escalating capacity demands. Specifically, to aggregate a higher number of optical channels, multiple parallel connections are routinely established between a muxponder and a Reconfigurable Optical Add-Drop Multiplexer (ROADM). A critical physical constraint in this architecture is that the same spectrum cannot be reused across parallel links that terminate at the same degree of a ROADM device without causing severe interference. Traditional RSA algorithms primarily focus on path computation, spectrum continuity, and spectrum contiguity across single-fiber logical links. However, they frequently fail to account for the spatial complexity introduced by multi-fiber or parallel-link topologies. Current RSA algorithms already face significant scalability and efficiency drawbacks in complex, highly meshed topologies; introducing parallel links exacerbates these challenges dramatically. In a multi-graph network featuring a uniform or variable number of parallel links between adjacent devices, the total number of possible path combinations from a source to a destination increases exponentially. Because signals routed over the same ROADM degree cannot share

frequencies, a network controller must exhaustively calculate and verify the availability of every frequency slot across every parallel edge. Computing paths in such multi-dimensional scenarios can easily overwhelm current algorithms, resulting in massive computational overhead, unacceptable latency, and ultimately, a failure in dynamic lightpath establishment. Consequently, there remains a critical research gap in developing routing strategies that can navigate the spatial complexity of parallel links without crippling network efficiency.

2.6 Current approaches and limitation

To address the NP-hard complexities of optical network management, theoretical research has developed sophisticated Routing and Spectrum Allocation (RSA) algorithms designed to mitigate severe spectrum fragmentation and maximize throughput. These strategies have evolved from foundational, decoupled wavelength assignment models and offline traffic sorting heuristics, into dynamic, multi-metric optimization approaches that actively steer traffic away from congested or unreliable paths. Recent literature also heavily emphasizes "next-state-aware" metrics, dynamic link-weighting, and complex defragmentation models. However, extensive taxonomic reviews reveal a prevailing structural limitation: the vast majority of these theoretical solutions are designed exclusively for one-dimensional, single-fiber-per-link topologies. When applied to multi-fiber scenarios, these models must treat parallel links either as physically invalid aggregated bandwidth or as entirely separate logical links, inevitably triggering a state-space explosion and unmanageable computational complexity.

In practical deployments, industry-standard SDN controllers like ONOS[23] and OpenDaylight (ODL)[24] attempt to manage these topologies using robust, graph-based routing engines, yet they fundamentally struggle with the same spatial constraints. In both controllers, optical connections are treated as strictly matched port-to-port links; thus, multiple parallel fibers between the same devices are mapped as entirely distinct and independent topological edges. When computing routes via standard K-Shortest Path algorithms, the topology engine outputs every permutation of these parallel links as a completely independent path sequence. Consequently, the subsequent spectrum allocation phase applies strict isolation, evaluating spectrum continuity and contiguity sequentially on a strict per-path basis. If an allocation attempt experiences wavelength contention on one parallel link, the controller discards the effort and must iteratively re-run the entire RSA verification from scratch on the next topological path permutation. In complex, highly meshed topologies featuring multiple parallel links, this exhaustive, sequential path-by-path iteration causes exponential computational overhead, often leading to unacceptable latency and dynamic provisioning timeouts. These systemic shortcomings in both theoretical models and practical SDN implementations highlight a critical research gap, presenting the opportunity to design a deterministic, spatially-aware parallel optical controller within the TeraFlowSDN ecosystem that evaluates parallel ports concurrently to

bypass sequential iteration and enable highly scalable lightpath establishment.

2.7 Related work

While the Routing and Spectrum Allocation (RSA) problem has been extensively explored to address strict spectrum continuity and contiguity constraints, existing literature and practical deployments exhibit a critical blind spot regarding multi-fiber topologies. Advanced theoretical heuristics—such as next-state-aware metrics and dynamic link-weighting—are overwhelmingly optimized for one-dimensional logical links. When extended to parallel-link environments, both theoretical models and industry-standard SDN controllers typically resort to exhaustive, sequential path-by-path iterations or restrictive offline computations, leading to severe latency and state-space explosion. To bridge this gap, this thesis introduces the **ParallelOpticalController** within the TeraFlowSDN ecosystem. By concurrently evaluating all available parallel ports at each hop rather than processing them as isolated topological sequences, the proposed architecture natively leverages the spatial domain to resolve RSA constraints in real time, bypassing sequential bottlenecks and enabling highly scalable, low-latency lightpath establishment for next-generation multi-fiber networks.

3 Problem Definition and Proposed Algorithm

This chapter formalizes the Routing and Spectrum Allocation (RSA) challenge within multi-graph optical topologies and presents the core algorithmic solutions developed for this thesis. It details the methodologies used to model physical networks with parallel links, compute required bandwidth and slots, discover optimal and alternative paths, and dynamically allocate spectrum using a novel bitmap intersection approach.

3.1 Formal Problem Statement

To address the complexities of lightpath provisioning in modern elastic optical networks, the physical topology must first be formalized into a mathematical model. The optical network is represented as a directed multi-graph, denoted as $G = (N, L)$. In this model, N represents the set of optical network nodes (e.g., transponders, muxponders, and ROADMs), and L represents the set of bidirectional physical optical links interconnecting these nodes. A defining characteristic of the targeted architecture is the presence of parallel links—multiple distinct physical fibers connecting the same node pair $(u, v) \in N$. Therefore, the edge set L is defined such that multiple edges can exist between any two vertices, formally expressed as a multi-graph where a link $l \in L$ is defined by the tuple (u, v, k) , with k serving as a unique identifier for a specific parallel fiber between nodes u and v . Each link l supports a total spectrum capacity divided into a set of frequency slots, W , adhering to standard ITU-T flexible grid specifications.

The network is governed by a centralized control plane, orchestrated by the TeraFlowSDN ecosystem. Within this environment, it is assumed that several active lightpaths, denoted as $LP_i(N_s, N_d)$, already exist, utilizing specific continuous and contiguous blocks of spectrum $w_i \subset W$ across various links to connect source nodes N_s to destination nodes N_d . The core challenge arises when a new lightpath request is submitted to the controller. The controller cannot simply compute the shortest path by treating the network as a standard simple graph. It must account for the current spectral state of the network, recognizing that certain links may be nearing capacity. Most critically, the routing algorithm must resolve the stringent physical constraints imposed by parallel links terminating at the same ROADM device. Specifically, in modern deployments, multiple parallel fibers originating from the same muxponder are frequently directed to the same Shared Risk Group (SRG)

and subsequently multiplexed into the same Degree (DEG) of a ROADM. Because these signals share the same physical amplification and switching infrastructure within the DEG, they are subject to strict spectral isolation constraints. If active lightpaths currently utilize a specific set of frequency slots on link $l(u, v, 1)$, those identical frequencies cannot be reused on a parallel link $l(u, v, 2)$ connecting the same devices, as doing so would result in severe optical interference. Therefore, when establishing a new lightpath across a parallel edge $l(u, v, k)$, the routing logic cannot evaluate the spectrum availability of link k in isolation. It must perform a concurrent evaluation of the entire spatial bundle of parallel links connecting u and v to ensure that the newly allocated contiguous spectrum block does not overlap with any frequencies currently active on sibling links i or j . The formal objective of the `ParallelOpticalController` is to efficiently compute the optimal routing path and allocate the necessary contiguous spectrum for incoming requests, specifically navigating the multi-dimensional constraints of parallel links, while minimizing overall network blocking probability and computational latency.

3.2 Graph Modelling with Parallel Links (topologies)

To efficiently navigate the multi-dimensional constraints established in the formal problem statement, the `ParallelOpticalController` avoids performing complex computations directly on the raw physical topology. Instead, the topological graph is logically segregated into specialized, abstracted versions. Acting as the centralized intelligence of the network, the TeraFlowSDN ecosystem provides the real-time database and telemetry required to construct, filter, and dynamically update these graphical models prior to processing any lightpath request. The core topology is first logically abstracted into an available-resource graph, denoted as G_{free} . The purpose of G_{free} is to immediately prune physical resources that cannot mathematically support a new connection, thereby shrinking the search space. During its construction, the controller queries the TeraFlowSDN database and actively excludes fully occupied Optical Multiplex Section (OMS) links between ROADMs, as well as currently utilized Optical Channel (OCH) ports between muxponders and ROADMs. The resulting G_{free} remains a multi-graph containing parallel links, but it strictly represents the network's actual residual capacity. While G_{free} accurately reflects available resources, executing shortest-path algorithms directly on a dense multi-graph introduces significant, often exponential, computational overhead. To bypass this state-space explosion, G_{free} is further abstracted into a simplified graph, denoted as G_{simple} . In this reduction, all parallel physical fibers connecting any adjacent node pair (u, v) are collapsed into a single representative logical edge. This transformation allows the controller to rapidly execute standard shortest-path and alternative-path computations (such as Dijkstra's shortest path algorithm) to discover the optimal topological routes from the source to the destination. Crucially, G_{simple} is used strictly for topological path discovery, not for spectrum allocation.

Once the candidate node-to-node routes are identified, the controller projects these paths back onto the multi-graph G_{free} to perform the actual Routing and Spectrum Allocation (RSA). By evaluating the physical routes against G_{free} , the controller natively re-introduces the spatial domain—allowing it to concurrently evaluate all parallel fibers for a given hop. This dual-graph methodology ensures that the path finding phase remains computationally lightweight, while the final RSA phase mathematically guarantees that utilized spectrum is accurately nullified and the contiguity constraints are reliably satisfied across parallel links.

3.3 Proposed RSA Algorithm

The proposed Routing and Spectrum Allocation (RSA) algorithm operates through a sequence of integrated stages to dynamically provision lightpaths. The process is initiated when a service request is submitted via the TeraFlowSDN WebUI using a JSON descriptor. The controller parses this descriptor to extract critical constraints, such as the requested bandwidth and the hop-count limit for alternative path computation. Following this extraction, the algorithm systematically executes bandwidth and slot calculation, topological path discovery, and spatial spectrum allocation analysis across parallel links. Finally, rather than forcing an automated assignment, the architecture presents the computed valid path permutations intuitively via the control plane, allowing the network operator to explicitly select the desired spatial route for final spectrum acquisition and assignment.

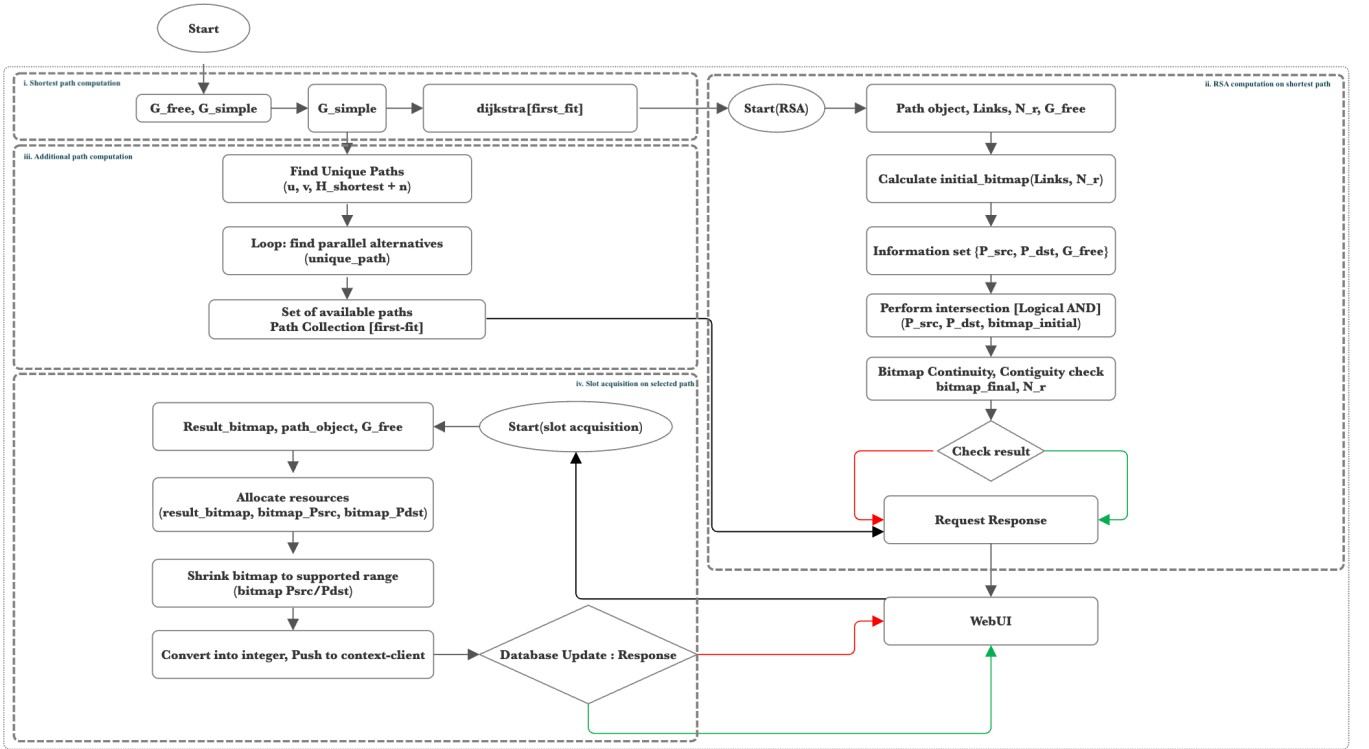


Figure 3.1: RSA operation using ParallelOpticalController in TeraFlowSDN

3.3.1 Bandwidth & Slot Computation

The initial phase of the proposed algorithm determines the required spectral bandwidth (B_r) for the requested optical service. Upon receiving a connection request, the controller extracts the target transmission bitrate (R_b), expressed in Gigabits per second (Gbps), directly from the submitted TeraFlowSDN service descriptor. The required bandwidth is then mathematically derived using the following formulation:

$$B_r = \frac{R_b}{n}(1 + \alpha)$$

where $n = \log_2(M)$, and M represents the specific modulation format utilized. While several factors influence the required bandwidth calculation [25], this standard formula simplifies the process for the controller.

Once the required bandwidth B_r is established, the controller computes the discrete number of frequency slots (N_r) needed to accommodate the lightpath. In accordance with ITU-T flexible grid standards, while 12.5 GHz typically represents the minimum routable spectrum width, the nominal central frequency granularity of 6.25 GHz is utilized as the fundamental unit for this computation. This finer 6.25 GHz granularity is deliberately adopted to streamline the mathematical modeling and facilitate the subsequent bitmap-based spectrum assignment logic. The required number of slots is calculated as follows:

$$N_r = \frac{B_r}{6.25}$$

Upon completion of this phase, the computed parameters, B_r and N_r , are temporarily cached within the controller's operational memory. These values serve as the primary spectral constraints for the ensuing topological path discovery and spatial spectrum allocation phases.

3.3.2 Path Discovery

Following the determination of the required bandwidth and frequency slots, the controller advances to the topological path discovery phase. To perform fast and computationally efficient routing, the algorithm leverages NetworkX[26], a robust and highly optimized Python library designed for complex network analysis. During this phase, the routing engine queries the simplified graph abstraction, G_{simple} —where parallel physical fibers are collapsed into single logical edges. By routing over G_{simple} rather than the raw physical multi-graph, the controller effectively prevents the state-space explosion and latency bottlenecks that plague traditional routing algorithms. The path computation engine is designed to execute two distinct routing operations to provide the user with maximum flexibility: it first identifies the absolute shortest path, and subsequently computes a set of viable alternative routes. This alternative path computation is governed by a user-defined hop-relaxation constraint, formulated as $H_{shortest} + n$, where $H_{shortest}$ represents the hop count of the primary shortest path

and n is a positive integer configured via the TeraFlowSDN service descriptor. The algorithm filters the topology to discover and validate alternative routes that strictly conform to this exact hop-count limit.

Once these node-level routes are successfully discovered in G_{simple} , the controller projects them back onto the available-resource multi-graph, G_{free} . Naturally, in a multi-fiber environment, a single node-to-node logical route expands into multiple physical path permutations between the source (N_s) and destination (N_d) devices. For standard operational deployment within the TeraFlowSDN ecosystem, the controller is configured to default to the first valid physical path object among these parallel permutations that satisfies the spectral constraints. However, to comprehensively evaluate the controller's spatial awareness, several alternate spatial selection strategies have been investigated, with their statistical impacts on network performance and blocking probability detailed in the subsequent analytical sections of this thesis.

3.3.3 Bitmap-Based Spectrum Allocation

Upon successful path discovery, the controller initiates the spectrum allocation phase using the primary shortest-path object derived from Dijkstra's algorithm, along with the previously computed slot requirement, N_r . While the controller defaults to evaluating the primary shortest path, RSA computation for any discovered alternative routes can be executed subsequently at the operator's discretion. The allocation process begins by querying the controller's internal cache to retrieve the real-time operational state of all physical links comprising the selected path object. The foundational step in this RSA methodology is establishing a uniform spectral reference framework. The controller analyzes the path to identify the maximum spectral band supported by any traversed device. Within the cache, the current spectrum utilization of each port is stored not just as discrete slots, but mathematically encoded as an integer value representing a binary sequence. This integer, referred to as the **bitmap-value**, is calculated using the following formulation:

$$\text{bitmap-value} = 2^N - 1$$

The integer representing the widest supported operational band (typically the C+L band) is designated as the **bitmap_initial**. This value establishes the maximum bit-length reference for all subsequent logical operations. A defining feature of this RSA algorithm is its native, unconditional sensitivity to multi-fiber topologies. Regardless of whether a specific hop explicitly utilizes multiple fibers, the algorithm inherently executes a parallelism check, ensuring that spectrum continuity and contiguity are dynamically evaluated across all available spatial dimensions between adjacent nodes. For a given hop along the path from source N_s to destination N_d , the controller aggregates all available source ports into a set P_{src} and all corresponding destination ports into a set P_{dst} . The current spectrum utilization of each port is retrieved as its integer **Bitmap_value** and converted into its binary representation.

To ensure mathematical alignment, each binary sequence is zero-padded or expanded to perfectly match the bit-length of the reference `bitmap_initial`. Once aligned, the controller executes a concurrent bitwise intersection (logical AND operations) across the expanded bitmaps of all eligible source and destination ports within that hop. This concurrent spatial evaluation collapses the multi-fiber complexity into a single, unified mathematical representation, yielding a definitive `bitmap_final`. This `bitmap_final` sequence explicitly maps the continuous and contiguous spectrum gaps that are simultaneously available across the entire parallel link bundle. The controller then applies a standard First-Fit policy to the `bitmap_final` sequence to identify the lowest-indexed contiguous block of zeros (available spectrum) that meets the required length N_r . The results of this deterministic spatial-spectral evaluation, including the final computed bitmap and the corresponding physical path hops, are subsequently relayed to the TeraFlowSDN WebUI for operator review.

3.3.4 Slot Acquisition & Assignment

The final phase of the proposed RSA algorithm transitions from computation to operational execution, facilitating the actual acquisition and assignment of spectral slots. Rather than autonomously enforcing an allocation, the architecture employs an interactive, operator-in-the-loop paradigm. The network operator is presented with the computed primary and alternative path objects—along with their corresponding available spectrum blocks—and is granted the flexibility to explore these options and explicitly select the preferred spatial route via the TeraFlowSDN WebUI. Upon operator’s confirmation, the controller initiates the slot acquisition protocol for the selected path. It retrieves the current link information for the chosen route and updates the spectral state by converting the newly allocated continuous blocks from their binary bitmap representations back into integer values. These updated integer values are subsequently pushed to the centralized database via the TeraFlowSDN Context component (e.g., using the Context client API) to accurately reflect the network’s newly diminished capacity.

Concurrently, the controller executes a series of state updates across the topology. At the central database level, the specific spatial device ports utilized by the lightpath are flagged as `used=True` to prevent future routing collisions. At the orchestration level, the computed lightpath object is transitioned to an "active" state, and the overarching service request is formally marked as "ACTIVE" within the central knowledge base. Operators can subsequently monitor these allocated resources, port usages, and path details through the WebUI. Crucially, within the current scope of this implementation, the `ParallelOpticalController` operates strictly as a control-plane and orchestration entity. While it comprehensively updates the centralized database and topological state to enforce resource locking, it does not currently generate or transmit southbound physical configuration commands (such as NETCONF or RESTCONF) to alter the hardware interfaces of the physical optical devices. All spectral acquisition and state management are logically enforced at the

controller level.

3.4 Complexity Analysis

The computational efficiency of the proposed **ParallelOpticalController** is fundamentally anchored in its ability to decouple topological routing from spatial spectrum allocation. The initialization phase, which constructs the abstracted multi-graph (G_{free}) and simplified routing graph (G_{simple}), traverses the network state in strictly linear time, $O(|N| + |L|)$ —effectively $O(n)$ relative to the total topology size (n represents a unit consisted of N, L)—ensuring predictable memory and processing scaling as physical devices and links are added. Subsequently, by executing topological path discovery exclusively on the collapsed graph G_{simple} , the controller avoids the state-space explosion typical of multi-graph environments, maintaining Dijkstra’s standard log-linear time complexity of $O(|L_{simple}| + |L \times \log L|)$ (where L_{simple} represents the links involved in the shortest-path). The most significant computational breakthrough, however, emerges during the Routing and Spectrum Allocation (RSA) phase.

Topology	Graph gen.	Shortest path com.	RSA comp.	Additional path comp.	Efficiency
Linear Single	0.0018	0.0012	0.0003	0.0009	-
Linear Parallel	0.0018	0.0016	0.0005	0.0009	87.50%
Butterfly Single	0.0014	0.0008	0.0004	0.0007	-
Butterfly Parallel	0.0017	0.0016	0.0007	0.0007	70.21%
NSF Single	0.0019	0.0016	0.0007	0.0023	-
NSF Parallel	0.0022	0.0021	0.0011	0.0024	83.33%
PANEU Single	0.0024	0.0024	0.0013	0.0134	-
PANEU Parallel	0.0026	0.0027	0.0019	0.0141	91.54%

Table 3.1: Efficiency evaluation of ParallelOpticalController

While traditional SDN implementations evaluate multi-fiber topologies sequentially—resulting in an exponential time complexity of $O(L_{parallel}^H)$, where $L_{parallel}$ represents the number of parallel links per hop and H denotes the path length(measured in hops)—the proposed algorithm evaluates spatial bundles concurrently. Because the bitwise intersection mathematical logic resolves spectral contiguity across all parallel ports in constant $O(1)$ time at the CPU hardware level, the overall time complexity of the RSA phase is strictly bounded by $O(L_{parallel} \times H)$. This additive, linear scaling with respect to path length and fiber density, coupled with a minimal spatial footprint of $O(N + L)$ for graph storage and negligible auxiliary memory for the binary string operations, mathematically guarantees that the controller can provision complex, multi-fiber lightpaths without succumbing to the exponential latency bottlenecks inherent in conventional routing algorithms. To empirically validate these theoretical claims, comprehensive performance tests were conducted across varying network topologies, including well-known reference networks such as the Pan-European (PANEU) and NSFNet models. As demonstrated in the results

below, the controller successfully maintains the mathematically established time and space complexities, exhibiting highly efficient scaling even when multi-fiber parallel links are introduced. Ultimately, the convergence of theoretical analysis and empirical performance metrics confirms the robust computational efficiency of the `ParallelOpticalController`. By leveraging concurrent spatial evaluation to eliminate sequential processing bottlenecks, the architecture ensures highly scalable, real-time lightpath provisioning capable of meeting the rigorous demands of next-generation, multi-fiber elastic optical networks.

4 System Design and Implementation

This chapter provides a detailed technical overview of the system architecture and the specific software implementations developed to support parallel optical links. It outlines the extensive modifications made to the TeraFlowSDN ecosystem, the creation of a standalone RSA computation module, and the deployment of the novel `ParallelOpticalController` microservice.

4.1 Overall System Architecture

The proposed solution is built entirely within the TeraFlowSDN (TFS) microservices ecosystem. TFS operates as a cloud-native, Kubernetes-based SDN controller, making it highly modular and extensible. To solve the Routing and Spectrum Allocation (RSA) problem for parallel links, the system architecture was augmented with a newly developed `ParallelOpticalController` service. This service interfaces heavily with the existing `ContextService` (for retrieving topology and endpoint database states) and the `DeviceService` (for communicating with the physical and emulated optical hardware). The architecture ensures that lightpath requests are intercepted by the `ParallelOpticalController`, which computes the optimal routing and spectrum allocation before instructing the `DeviceService` to push XML configurations to the underlying transponders and ROADMs.

4.2 Modifications to TeraFlowSDN

4.2.1 Device Service Enhancements (`OCDriver`, `OpenROADMDriver`)

To perform accurate device-based spectrum filtration, the controller requires detailed hardware telemetry. The existing `OCDriver` (for OpenConfig devices) was heavily modified to extract device information directly from the device's XML configuration files rather than relying solely on static JSON descriptors.

1. A new device-specific filter mechanism was introduced, establishing a `device_filters` package containing classes like `DefaultDevice.py` and `CNITTestROADMDevice.py`.

2. The `OCDriver.py` was updated to execute a two-phase extraction process: it first retrieves the `model_name` from the device, and then maps this model to the appropriate filter class to extract transponder and port values accurately.
3. Similar enhancements were made to the `OpenROADMDriver`, filtering out internal mapping ports and correctly marking client ports as OCH transport types and line ports as OMS transport types.

4.2.2 Context Service & Database Schema Extensions

Because the original database lacked fields for manufacturer and hardware specifications, significant schema extensions were introduced to `ContextService`.

1. The `DeviceModel.py` table was updated to include mandatory fields: `mfg_name`, `model`, and `serial_no`, alongside nullable fields for `hardware_version` and `software_version`.
2. The `EndPointModel.py` was extended to include an index string and a `transport_type`. The index acts as a unique identifier (e.g., "TP1_11") generated by combining a sanitized device name and the port index, resolving previous endpoint mapping ambiguities.
3. The `OpticalChannelModel` was upgraded to store crucial RSA metrics: `min_frequency`, `max_frequency`, `flex_slots`, and `bitmap_value`. These schema changes were formally defined in the `context.proto` protocol buffers and propagated throughout the gRPC services.

4.3 RSA Computation Module (RSATools, ITUStandards)

To centralize the mathematical and logical rules of optical networking, two primary Python helper modules were developed: `ITUStandards.py` and `RSATools.py`.

- **ITUStandards.py:** This module codifies the strict ITU-T G.694.1 standards. It defines constants such as the 193.1 THz anchor frequency and the 6.25 GHz slot granularity. It also encapsulates enumerations for frequency ranges across all optical bands (O, E, S, C, L) and slot availability statuses (AVAILABLE, UNAVAILABLE, IN_USE).
- **RSATools.py:** Functioning as the core logic engine, this module handles the dynamic RSA calculations. It includes functions like `detect_band()` to identify the smallest optical band containing a given frequency, and `compute_rsa_params()` to determine the minimum and maximum frequencies and required flex slots. Crucially, it calculates the base hardware spectrum bitmap

using the formula $B_{\text{map}} = 2^{\text{slots}} - 1$ and dynamically generates spectrum slot tables for both routing computations and WebUI visualization.

4.4 Parallel Optical Controller Microservice

4.4.1 Service Architecture & API Design

The ParallelOpticalController was engineered as an autonomous microservice within TFS. It is containerized using Docker and deployed into the Kubernetes cluster alongside existing services. Following the deployment, both the serviceservice and webuiservice undergo rollout restarts to discover the new controller's environment variables and navigation endpoints.

4.4.2 Graph Construction with NetworkX

When an optical intent is received, the controller does not rely on static routing tables; instead, it dynamically reconstructs the network graph. Fetching endpoint and optical link data via the Context client, it utilizes the networkX package to build complex multi-directed graphs. The module `topology.py` is responsible for defining graph nodes (active optical devices) and generating distinct graph models, such as `G_FREE` for links where OCH is unused, and `G_simple` for raw path discovery.

4.5 Integration with Existing TFS Services

The controller interfaces seamlessly with the ContextService to read the current network state. During the "Slot Acquisition" phase, the controller updates the flow parameters in the database via the Context client, ensuring the allocated bitwise spectrum blocks are officially locked and recorded.

4.6 Topology & Link Management (Parallel Link Support)

A major implementation hurdle was supporting topologies with parallel links natively. The traditional method of mapping links via `endpoint_uuid` often failed when deploying topologies dynamically due to hashing mismatches. To resolve this, link descriptors were upgraded to support an `endpoint_index` fallback mechanism. Furthermore, to maintain a clean graphical user interface without overlapping link lines, parallel links are grouped and visually represented using an integer multiplier format, such as `xN`, detailing the exact number of parallel connections existing between two adjacent devices.

4.7 WebUI Enhancements (Spectrum Visualization)

To expose the advanced capabilities of the backend, the TFS Web User Interface underwent significant improvements.

- The device details template (detail.html) was modified to include the newly extracted Index and Transport Type metrics.
- A dedicated "Endpoint Frequency View" template was engineered, featuring a comprehensive "Frequency Capabilities Card" that displays wavelength ranges, bandwidth, and the specific optical band name.
- Most notably, the WebUI now features a real-time, scrollable "Spectrum Allocation Table". This table displays every individual 6.25 GHz slot within the device's band, utilizing color-coded badges to indicate slot status: green for AVAILABLE, yellow for IN_USE, gray for UNAVAILABLE (outside hardware bounds), and blue for the standard ANCHOR frequency.

5 Testbed Setup and Deployment

This chapter outlines the practical environment established to validate the proposed Routing and Spectrum Allocation (RSA) algorithms and the ParallelOpticalController. It details the configuration of the physical servers, the initialization of the Docker networking layer, the setup of emulated optical devices via XML data models, and the final microservices deployment procedure utilizing Kubernetes (MicroK8s).

5.1 Physical/Virtual Infrastructure

The experimental testbed relies on a distributed server architecture hosted at the University of Pisa (UniPi). Two primary servers, designated as Monster and Mascara, are utilized to balance the computational load between the SDN control plane and the emulated data plane.

To facilitate communication between the TeraFlowSDN controllers and the emulated devices, an isolated Docker bridge network is established. The creation of this network on the Monster server requires specifying a dedicated subnet and IP range:

```
1  # On secondary server
2  sudo docker network create --driver=bridge \
3  --ip-range=10.100.0.0/16 \
4  --subnet=10.100.0.0/16 \
5  -o "com.docker.network.bridge.name=brTFS" netbrTFS
```

Listing 5.1: Creating docker network

To ensure the Mascara server can route traffic to this newly created Docker subnet, static IP routing is configured. The following commands define the route through the respective gateway IP:

```
1  # Executed on Primary Server
2  sudo ip route add 10.100.0.0/16 via 131.114.54.73
3  route -n # Verification of the routing table
```

Listing 5.2: Creating route to the docker network

Furthermore, due to operating system constraints on Ubuntu 22.04 regarding network persistency, explicit iptables rules must be inserted on the Monster server to accept incoming traffic from the Mascara node:

```
1  # Executed on UniPi Server: Monster
2  sudo iptables -I DOCKER-USER -s 131.114.54.72 -d 10.100.0.0/16 -j ACCEPT
3  sudo iptables --list # Verification of firewall rules
```

Listing 5.3: Adding exceptions for teraflow packets

[This section will be written properly, because I have to introduce two setup. One for debian system with older versions of docker and the new one with ubuntu 22.04 and latest version of docker.]

5.2 Emulated Devices (Transponders, ROADMS)

Because deploying physical, large-scale flex-grid optical hardware is cost-prohibitive, the data plane is entirely simulated using advanced device emulators. These emulators accurately replicate the NETCONF/YANG interfaces of commercial optical equipment.

The configuration of these devices relies heavily on meticulously crafted XML files that define their hardware capabilities, physical ports, and operational frequencies.

- **Transponders (Muxponders):** The transponder configurations utilize the OpenConfig standard. For example, the `transponders_x4.xml` file defines a MellanoxSwitch device with multiple `QSFP56_DD_TYPE1` transceivers operating under the `ETH_400GBASE_ZR` standard. It explicitly details the optical channels, defining parameters such as the default 193.1875 THz frequency, target output power, and associated line ports.
- **ROADMs:** Reconfigurable Optical Add-Drop Multiplexers are configured to simulate specific optical degrees and frequency bands. The `roadm.xml` file dictates the input and output directional mapping of ports (e.g., ports 4101 and 5201 mapping to Degree D1). Crucially, it hardcodes the hardware bounds for the proposed RSA algorithm, explicitly defining the `MG_ON_PORT_MIN_FREQUENCY` as 193.0 THz and the `MG_ON_PORT_MAX_FREQUENCY` as 193.375 THz.

5.3 Network Topology Configuration (tfs descriptors)

To push the network topology into the TeraFlowSDN Context database, standardized JSON descriptors are utilized. These descriptors dictate the node identities, endpoint indices, and the optical links connecting them. To support the testing of topologies with multiple parallel links, the topology descriptor parsing mechanism was updated. Instead of relying strictly on unstable UUID hashing, the testbed utilises descriptors that reference the deterministic `endpoint_index` (e.g., `TP1_11`), allowing stable, repeatable topology generations with parallel edge definitions (e.g., xN multiplier configurations).

5.4 Deployment Procedure (TeraFlowSDN & ParallelOpticalController)

The entire TeraFlowSDN ecosystem, including the custom-built components, is deployed on a MicroK8s Kubernetes cluster. When deploying the novel ParallelOpticalController to a fresh TFS ecosystem, a strict five-step deployment pipeline is executed .

First, the Docker image for the new controller is built and pushed to the local container registry:

[Deployment of TeraFlowSDN]

6 Results and Evaluation

This chapter presents the performance evaluation and functional validation of the proposed Routing and Spectrum Allocation (RSA) framework. It assesses the operational success of the newly integrated ParallelOpticalController within the TeraFlowSDN (TFS) ecosystem and analyzes the efficiency of the proposed bitwise spectrum allocation algorithm across topologies featuring parallel optical links.

6.1 Evaluation Methodology & Test Scenarios

To rigorously evaluate the system, a series of test scenarios were executed within the established Docker-based emulated environment. The evaluation methodology focuses on injecting dynamic optical intents (lightpath requests) into the TeraFlowSDN controller and monitoring the system’s response across multiple metrics.

The test scenarios utilize various network topologies, specifically focusing on multi-directed graphs where nodes (e.g., transponders and ROADMs) are connected via multiple parallel links, denoted by the xN multiplier format. The primary objectives of these tests are to validate the functional correctness of the device telemetry extraction, measure the computational efficiency of the path discovery algorithm, and statistically analyze the spectrum utilization under varying traffic loads.

TBD: (Detail the specific parameters of my test scenarios here, such as the exact number of nodes, the maximum number of parallel links per edge, the distribution of traffic requests, and the requested bandwidth/modulation formats used during the simulation.)

6.2 Functional Validation

Before analyzing quantitative performance, the functional correctness of the system was validated through backend logs and the enhanced Web User Interface (WebUI).

First, network connectivity and device emulation stability were verified by successfully pinging the emulated optical devices directly from the routing server (e.g., pinging 10.100.101.1 from the Mascara server). Following successful device discovery, the modified OCDriver and OpenROADM drivers successfully extracted the vendor-agnostic hardware parameters, which were confirmed via the Context Service database logs.

Furthermore, the functional validation is heavily supported by the newly developed visual representations in the TFS WebUI. When navigating to the device details page,

the system successfully renders the "Endpoint Frequency View" template. This view dynamically displays the accurate Index and Transport Type of each port, alongside a comprehensive "Frequency Capabilities Card" outlining the specific optical band, wavelength range, and total flexible-grid slots.

Crucially, the "Spectrum Allocation Table" provides a real-time, color-coded visual validation of the RSA algorithm's output. The table accurately reflects the status of individual 6.25 GHz slots:

- Gray badges correctly identify UNAVAILABLE slots that fall outside the physical device's capabilities.
- Green badges represent AVAILABLE slots ready for provisioning.
- Yellow badges indicate IN_USE slots successfully allocated by the ParallelOpticalController.
- Blue badges accurately mark the ITU-T standard ANCHOR frequency (193.1 THz).

6.3 Computational Performance

TBD: (This section will present the execution time and computational overhead of the ParallelOpticalController. I must include graphs or tables showing the time taken to construct the NetworkX graph, the time taken to run Dijkstra's and the `all_simple_paths()` algorithms, and the speed of the bitwise spectrum intersection process as the number of parallel links increases.)

6.4 Spectrum Utilization Efficiency

TBD: (This section will evaluate how efficiently the flexible-grid spectrum is packed. I will discuss the fragmentation of the optical bands and provide metrics on how the proposed algorithm maximizes the use of 6.25 GHz slots across complex topologies compared to traditional fixed-grid or sequential parallel-link checking methods.)

6.5 Throughput Analysis

TBD: (This section will analyze the network's overall throughput based on the blocking probability. I must present data comparing the volume of successfully allocated optical intents versus rejected requests due to spectrum continuity constraints. Include an analysis of network-level

efficiency by comparing the ratio of allocated slots to the total available slots across all active links.)

6.6 Comparison with Baseline/Existing Approaches

TBD: (Here, I will benchmark your ParallelOpticalController against the default, sequential RSA mechanisms found in standard SDN controllers (like ONOS, OpenDayLight or the baseline TeraFlowSDN OpticalController). Provide comparative charts highlighting improvements in path computation time, reduction in blocking probability, and overall algorithmic efficiency when handling the xN parallel link scenarios.)

6.7 Statistical Analysis

TBD: (This section requires a deeper mathematical and statistical evaluation of the results. I will provide an analysis of the blocking paths and theoretical channel capacity against the actual used capacity during peak simulation periods.)

6.8 Discussion of Results

TBD: (I will provide a comprehensive summary and interpretation of the findings from the previous sections. Discuss whether the research objectives were met, highlight the most significant improvements brought by our parallel RSA algorithm, and acknowledge any unexpected behaviors or minor anomalies observed during the testing phase.)

7 Conclusion and Future Work

This concluding chapter summarizes the overarching achievements and technical contributions presented throughout this thesis. It critically reflects on the practical constraints and structural limitations encountered during the implementation phase and outlines potential avenues for future research to further refine the management of flexible-grid optical networks within TeraFlowSDN.

7.1 Summary of Contributions

As the demand for high-capacity telecommunications infrastructure continues to grow, optimizing the allocation of optical spectrum across complex topologies has become paramount. This thesis directly addressed the computational bottlenecks associated with Routing and Spectrum Allocation (RSA) over parallel optical links—a critical area often overlooked or inefficiently handled by traditional Software-Defined Networking (SDN) controllers.

The primary contribution of this research was the successful design, development, and deployment of the `ParallelOpticalController`, a stateful microservice fully integrated into the TeraFlowSDN (TFS) ecosystem. By leveraging the `NetworkX` library to construct multi-directed graphs, the controller natively supports and efficiently routes optical intents across topologies featuring multiple parallel edges. To resolve the heavy computational burden of high-resolution spectrum mapping, the system introduced a novel bitmap-based intersection algorithm capable of rapidly computing the availability of contiguous 6.25 GHz slots across multiple hops.

To ground this algorithmic advancement in practical hardware realities, this work significantly enhanced the TeraFlowSDN `DeviceService` and `ContextService`. The `OCDriver` (`OpenConfig`) and `OpenROADM` drivers were upgraded with custom model-based filters to extract vendor-agnostic telemetry data—such as operating frequencies, transport types (OCH, OMS), and hardware specifications—directly from XML configurations.

Finally, these backend capabilities were made accessible through a comprehensive upgrade to the TeraFlowSDN WebUI. The introduction of the dynamic "Spectrum Allocation Table" provides network administrators with real-time, color-coded visual feedback regarding the status (e.g., `AVAILABLE`, `IN_USE`, `UNAVAILABLE`) of individual ITU-T G.694.1 slots, drastically improving overall network transparency and operational user experience.

7.2 Limitations

TBD: More sophisticated device-based filtration, stateful characteristics of the controller, comparatively slower DB operations with intense topological intents ,generating a reply with the results and update the device with a NETCONF request.

7.3 Future Work Directions

TBD

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